3'end RNA-seq using template switching protocol

Status: `[?]` **| partially migrated / unconfirmed**

Status legend: **[OK]** working `[?]` unconfirmed / partial [X] broken

About

Protocol for 3' end RNA-seq using a template switching reverse transcription workflow.

Contents

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1. RNA extraction

Input / starting material: RNA lysate

- Maxwell RNA extraction
- Alternative: Add 3 volumes of LS-Trizol to lysate
- Extract RNA using the Zymo MicroSpin Columns
- Elute with 6 µL of H O

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2. RNA fragmentation and template switching RT

- Prepare fragmentation buffer.
- Fragmentation buffer: 80 mM Tris-HCl pH 7.5, 40 mM MgCl . > **Note**: have more recently been using milder fragmentation buffer conditions with input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl (final concentration 1.25 mM Tris-HCl and 0.625 mM MgCl).
- Use 500 ng of RNA (whenever possible).
- Bring to 3.5 µL using water.

- Add 0.5 µL of fragmentation buffer (10 mM Tris-HCl pH 7.5, 5 mM MgCl final concentration).
- Fragment RNA by incubating 12 min at 95 °C.
- Add:
  - 0.5 µL 10 mM dNTPs
  - 0.5 µL 5 µM primer RT1
- Mix and incubate at 65 °C for 3 min then cool down to 42 °C and hold.

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3. Template switching RT incubation and cleanup

- Make a mastermix and add 5.5 µL of the following:
  - 2 µL of SSIV buffer
  - 2 µL 5 M betaine
  - 0.5 µL 0.1 M DTT
  - 0.25 µL 40 µM TSO
  - 0.25 µL 200 mM MgCl
  - 0.25 µL SuperRNasin
  - 0.25 µL SSIV
- Place the mastermix into a PCR strip and into the cycler so that it pre-warms to 42 °C before you start adding it to the samples.
- Add mastermix to samples while keeping them at 42 °C in the cycler (i.e., don't take the samples out of the cycles while adding the mix).
- Incubate at 42 °C for 10 min.
- Incubate at 50 °C for 30 min.
- Purify using Magbind beads (1.8 µL per 1 µL of PCR reaction). Elute in 8 µL H O.

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4. Optional PCR test

- Use 1 µL RT product.

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5. Indexing PCR

- Prepare the following 20 µL indexing PCR reaction for each sample:
  - 10 µL of 2× Q5
  - 2 µL RT product
  - 1 µL dual indexed primers
    - Use a different index for each sample.

        * Make sure each sample will have a different index if submitting multiple libraries at the same time.
    - 7 µL of ddH2O
- Incubate for the PCR reaction using the Q5 program:
    - Extension time – 45 seconds
    - Tm – 65 °C
    - Typically try 10 and 14 cycles
- Move samples to post-PCR room.
- Add 2 µL of purple 6X loading buffer to 5 µL of PCR reaction. Keep the rest of the 20 µL.
- Load on a 6% TBE gel together with 2 µL of ready-to-use low molecular weight ladder (NEB).
- Run at 180 V for 25 min.
- Stain with Sybr Green (1:10000 dilution) in 1X TBE.
- Visualise.

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6. PCR beads purification

- Mix together approx. 10 µL of each PCR reaction.
- Add a suitable amount of Magbind beads (e.g. 0.9 µL per 1 µL of PCR reaction) and mix.
- Incubate at room temperature for 5 minutes.
- Remove supernatant and wash with 500 µL of 70% Ethanol for 30 sec. Repeat this step.
- Air dry the beads.
- Resuspend in 30 µL of H O.

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7. Buffers

- Fragmentation buffer: 80 mM Tris-HCl pH 7.5, 40 mM MgCl .
- Milder fragmentation buffer conditions: input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl (final concentration 1.25 mM Tris-HCl and 0.625 mM MgCl).
